

CLASP: A Cluster-Level Autonomous Selective Picking Robot with a Soft Rolling-Band Gripper for Fresh-Market Blueberry Harvesting

Yixuan Xia*, Yilin Cai*, Natalia Belen Espinoza, Changying Li, Zilfina Rubio Ames, Xin Zhang, Yue Chen

Abstract—Fresh-market blueberries require selective, gentle picking, which is labor-intensive and expensive. Over-the-row machine harvesters are fast but non-selective, bruising mixed-ripeness fruit and limiting yield to the processing market. Selective robotic harvesters typically target individual fruits rather than fruit clusters, which limits harvesting efficiency for small, densely clustered blueberries. This paper presents CLASP, a Cluster-Level Autonomous Selective Picking robot with a Soft Active Rolling-Band Gripper (SARB-Gripper). Two compliant bands envelop the cluster and roll against the fruit, drawing mature berries off in sequence, while closed-loop regulation of the pulling force keeps the applied load below the immature detachment threshold. A global-to-local perception pipeline pairs an eye-to-hand camera for global cluster detection and target selection with an eye-in-hand camera for local localization and cluster orientation estimation. Field measurements confirm a clear detachment-force separation between mature and immature fruit, and the SARB-Gripper reproduces a commanded pulling force to within 3.7 %, enabling selective harvesting at the cluster level. In end-to-end field trials, CLASP autonomously grasped 23 of 25 presented clusters (92 %). With the component cost of approximately \$3326 per unit, CLASP offers a scalable approach to selective cluster-level harvesting for fresh-market blueberries.

Index Terms—Grippers, Agriculture Robot, Cluster-level Blueberry Harvesting

I. INTRODUCTION

U.S. cultivated blueberry production reached about 358,000 tons with a farm-gate value of roughly \$1.15 billion in 2024 [1]. Yet harvesting for the fresh market remains almost entirely manual [2], because within a single cluster blueberries ripen asynchronously. Pickers must pass through the field repeatedly, harvesting only ripe fruit while leaving the rest to

mature. Harvesting is thus the most labor-intensive and costly stage of fresh-market production, and as seasonal farm labor grows scarce, the pressure to automate it intensifies.

Despite this strong need for automation, fresh-market blueberry harvesting presents three coupled challenges. The first is *gentle handling*: conventional over-the-row harvesters shake blueberries from the bush, but studies show that mechanically harvested berries exhibits 16 % to 26 % severe bruising even with soft-catch surfaces, compared with only 1 % to 4 % for hand-harvested berries [9]. This damage confines shaking largely to the processing market and leaves its fresh-market economics marginal [10]. The second is *ripeness selectivity*: ripe and unripe berries are intermingled on a common stem, so a harvester must remove mature fruit while leaving immature fruit attached for later passes. The third is *dense cluster geometry*: blueberries are small and grow in tight clusters, where adjacent fruit contact and occlude one another. This structure makes it difficult to perceive and manipulate individual berries independently, limiting the applicability of per-fruit harvesting strategies developed for larger, sparsely distributed crops. Blueberry harvesting must therefore operate at the cluster level, while selectively detaching mature fruit and minimizing mechanical loading on the retained and harvested berries.

These three requirements are coupled, yet existing autonomous harvesting systems do not address them simultaneously. Most robotic harvesting systems developed for larger, sparsely distributed fruit follow a per-fruit paradigm, in which each target is individually localized and manipulated by the robot [11], [3], [4]. Meanwhile, compliant and soft grippers are developed to handle fruit gently [12], but are likewise typically designed to grasp one individually localized fruit at a time. This approach does not readily extend to blueberries, where harvesting must instead operate on multiple fruit within a cluster. Such cluster-level interaction introduces an additional requirement for selectivity, in which the applied load must be sufficient to detach mature berries while allowing immature berries to remain attached. The key challenge is therefore to move beyond individual-fruit manipulation toward a cluster-level operating principle that achieves selective detachment through controlled force rather than precise berry-level interaction.

To address these challenges, we present **CLASP**, a **Cluster-Level Autonomous Selective Picking** robot for fresh-market blueberry harvesting, shown in Fig. 1. Rather than localizing and manipulating each berry individually, CLASP uses a soft active rolling-band gripper (SARB-Gripper) to engage multiple berries simultaneously and progressively detach them through continuous rolling contact. Its pulling force

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TABLE I
INTEGRATED ROBOTIC HARVESTING SYSTEMS: TARGET CROPS, MANIPULATOR CLASS, COST TIER, AND REPORTED FIELD PERFORMANCE.

Crop	Manipulator	Class (cost tier)*	Success (%)	Cycle (s)	Environment
Apple [3]	Multi-arm [†]	Purpose-built	71.3–80.5	5.8–6.7	Orchard
Apple [4]	Custom 4-DOF × 2	Purpose-built	80.7 / 79.7	5.97	Orchard × 2
Strawberry [5]	Custom dual-arm	Purpose-built	50–97.1 ^a	4.6–6.1 ^b	Polytunnel
Strawberry [6]	Franka Panda 7-DOF	Commercial	83 ^c	—	Glasshouse, tunnel
Tomato [7]	5-DOF [†]	Commercial	~80	24.34	Laboratory
Pumpkin [8]	Custom, tractor-mtd	Purpose-built	92	35.1–58.7	Field
Blueberry (ours)	Custom 6-DOF	Purpose-built (\$3,326)	92^d	—	Open research field

*Tiers inferred from manipulator class where cost is unreported; [†]Model not stated in abstract. ^aFirst attempt, growth-situation dependent. ^bManipulation only. ^cOf fruits judged pluckable. ^dOf berry clusters presented to the system, counted as a success when the cluster is autonomously grasped.

is regulated using servomotor current feedback without a dedicated force/torque sensor. By maintaining the applied force within the detachment-force margin between mature and immature berries, the gripper selectively removes only the mature berries. Autonomous harvesting is enabled by a global-to-local perception and control pipeline. An eye-to-hand camera provides coarse cluster localization, while an eye-in-hand camera refines cluster geometry and assesses harvest completion. A finite-state controller coordinates perception, arm motion, gripper force regulation, and harvesting actions.

The main contributions of this work are:

- a cluster-level harvesting platform combining a compliant rolling-band gripper with sensorless force control for gentle, ripeness-selective detachment;
- a multi-season field dataset, together with a global-to-local perception pipeline for cluster detection, localization, orientation estimation, and harvesting progress monitoring;
- an integrated autonomous harvesting system validated under field conditions, achieving a 92 % cluster-level harvesting success rate.

II. RELATED WORKS

A. Manipulation Hardware for Selective Fruit Harvesting

Selective harvesting in dense canopies imposes two key requirements on robotic hardware: sufficient manipulator workspace and dexterity at an affordable cost, and compliant interaction between the end-effector and fruit. Most existing fruit-harvesting robots rely on commercial industrial arms, which provide the required dexterity and positioning accuracy [13]. However, their high cost limits the deployment in multi-arm systems, particularly for small farms [14]. Lower-DoF manipulators offer a more compact and cost-effective alternative, but typically provide a smaller workspace and reduced orientation dexterity [5], [8]. These capabilities are critical for accessing fruit targets in occluded canopies, where the robot must approach clusters from feasible directions while avoiding surrounding branches and foliage.

At the fruit interface, the end-effector must accommodate the delicate and irregular geometry of berry clusters. Compliant and soft grippers can reduce fruit damage and compensate for geometric uncertainty, realized via tendon-driven designs [15], [16], and suction-based designs [17], or linkage-based hybrid design [18]. However, these end-effectors primarily use compliance to facilitate gentle handling, without controlled force regulation for maturity-selective detachment.

B. Perception in Cluttered Agricultural Environments

For autonomous harvesting, successful fruit detection alone is insufficient. Perception must also provide sufficiently accurate spatial information to guide manipulation. Vision-based harvesting systems commonly combine learning-based detection and segmentation, including models from the YOLO-family, with RGB, stereo or RGB-D sensing feedback [19]. In field environments, however, harvesting performance is often constrained by occlusion, canopy structure, and target accessibility in addition to detector accuracy. For example, an apple harvesting pipeline reported a decrease in success rate from 82.4 % in a well-pruned orchard to 65.2 % in a dense orchard without changing the detector [11].

These constraints are particularly pronounced for blueberries. Prior blueberry perception studies have focused primarily on maturity classification, detection, and counting [20], [21], with recent work extending to 3D localization [22]. However, these methods have not been integrated into closed-loop robotic harvesting systems. For an RGB-D camera, fruit-center localization RMSE grows from 3.13 mm at a capture distance of 30 cm to 24.72 mm at 150 cm [23], exceeding the alignment tolerance of a berry-scale gripper [24]. These limitations thus motivate a global-to-local strategy that localizes the cluster globally and refines its local pose near the gripper.

C. Integrated Robotic Harvesting Systems

The design of an integrated robotic harvesting system depends strongly on the characteristics of the target crop. Integrated autonomous harvesting is established for larger separable fruits, including apples [11], [3], [4], strawberries [6], [25], mango [26], and tomato [7]. These systems typically follow a single-fruit picking workflow in which each target is localized, approached, and detached individually, at cycle times of roughly 6 s to 25 s and success rates of 70 % to 90 % (Table I). Cluster-level blueberry harvesting introduces different system-integration requirements. First, selectivity shifts from perception to the end effector, which must discriminate mature from immature fruit through the contact itself. Second, harvesting a cluster may require multiple detachment actions, requiring the system to monitor harvesting progress and determine when to proceed to the next target. To our knowledge, no field-evaluated robotic system integrates cluster-level targeting, selective detachment, and completion detection for fresh-market blueberry harvesting.

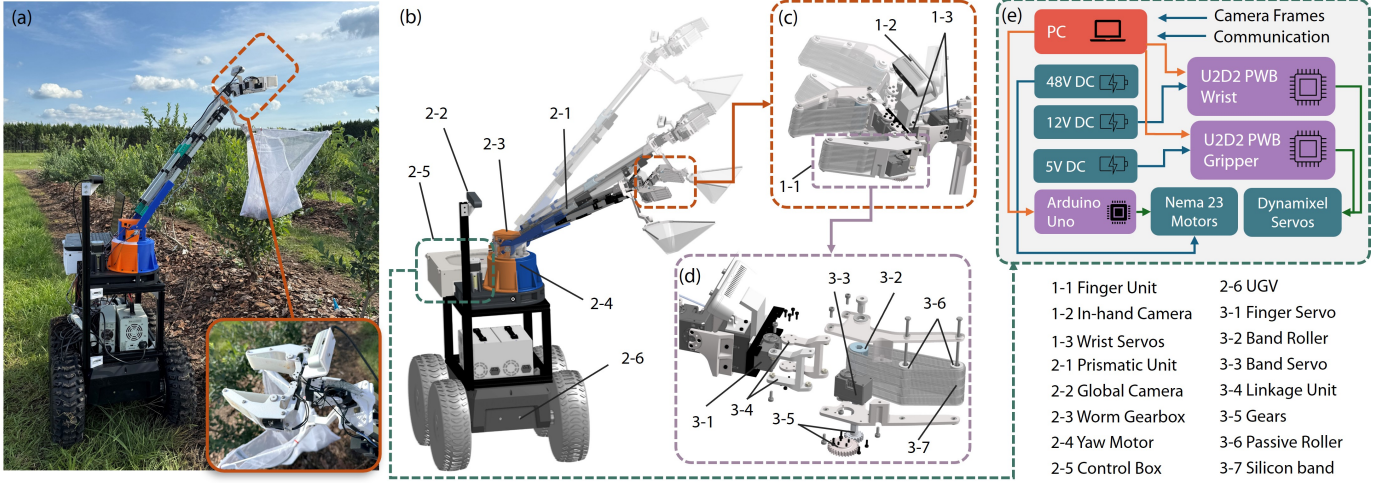


Fig. 1. Hardware design of the blueberry harvesting system: (a) system prototype in the field; (b) CAD model of the robotic system; (c) gripper design with three-DOF wrist; (d) exploded view of the finger unit; (e) electronic system.

III. MECHATRONIC DESIGN OF THE CLASP PLATFORM

A. Design of the 6-DOF Manipulator

The manipulator was designed to provide canopy-scale reach and local end-effector dexterity at low hardware cost. CLASP therefore uses a six-degree-of-freedom (6-DoF) serial manipulator with an RRP₄RR architecture, in which an extension joint is placed third in an otherwise revolute chain. The layout is shown in Figs. 1(a) and (b), and the corresponding Denavit–Hartenberg parameters are listed in Table II.

Actuation is split between stepper and servo motors. The azimuth joint is driven by a NEMA 23 stepper motor through a 47:1 gearbox and a timing belt. The elevation joint uses a NEMA 23 stepper motor with a 40:1 vertical worm gearbox. The self-locking transmission maintains the arm configuration without continuous holding torque and prevents uncontrolled motion under power loss. Downstream of these two joints, a Dynamixel XC430 servo drives the extension joint through a timing belt. The radial reach extends from 937 mm to 1333 mm. The wrist provides roll, pitch, and yaw through one Dynamixel 2XC430 dual-axis servo and one Dynamixel XC430 servo. Joints θ_4 , θ_5 , and θ_6 rotate about the wrist frame x , y , and z axes respectively, with z along the approach direction. The complete platform was constructed at a total component cost of approximately USD 3326.

Actuator selection follows the load distribution along the kinematic chain. The two proximal joints carry the full arm inertia. Steppers provide the required torque at lower cost than comparable servo-gearbox units. The AS5600 encoders recover the position feedback that open-loop steppers lack. The four distal joints move only the wrist and end-effector. Lower-torque Dynamixel servos are sufficient there and return position, velocity, and current feedback on a single bus. The extension joint decouples radial extension from the revolute chain. The arm therefore spans the canopy without the long links and high holding torques of an all-revolute design, and the wrist retains the approach orientations required in an occluded canopy.

TABLE II
DENAVIT–HARTENBERG PARAMETERS AND JOINT LIMITS OF THE 6-DOF MANIPULATOR (NOMINAL CAD VALUES)

Joint	θ_i (deg)	d_i (mm)	a_i (mm)	α_i (deg)	Range (deg)
Azimuth joint	θ_1	232.78	0	-90	$[-90, 65]$
Elevation joint	θ_2	0	-36.69	90	$[-15, 90]$
Extension joint	90	$764.25 + d_3$	0	90	$[0, 396]$
Wrist roll	$\theta_4 - 90$	0	-24	90	$[-30, 30]$
Wrist pitch	$-\theta_5 - 90$	0	0	90	$[-60, 20]$
Wrist yaw	$\theta_6 + 180$	148	0	0	$[-45, 45]$

B. Design of the SARB-Gripper

The Soft Active Rolling-Band Gripper (SARB-Gripper) draws inspiration from the motion used by human pickers, who use the thumb to gently pull or rub berries against the supporting stem to detach them, while the palm is positioned underneath to catch the detached fruit (Fig. 2(a)). The thumb contacts several berries at once and does not target them individually. The ripe berries separate under applied load, while the unripe berries remain attached. This picking strategy motivates two coupled functions in the gripper: compliant engagement with the cluster and active tangential motion for fruit detachment.

As shown in Fig. 1(c), the gripper consists of two fingers, each equipped with a soft rolling band made of Dragon Skin 20 silicone. Each band is independently actuated by a Dynamixel XC330 servo through a 0.6:1 gear transmission. The two fingers are driven by separate Dynamixel XC330 servos through four-bar linkages (Fig. 1(d)). The gripper provides a maximum finger opening of 100 mm, and a band contact length of 100 mm. During harvesting, the fingers are first closed around the target cluster to establish compliant contact. Rotating the bands drives relative sliding at the fruit–band interface, so interfacial friction detaches the contacted berries at the berry–pedicel junction (Fig. 2(b)). The detached berries fall into a soft net stretched between two aluminum rods below the gripper. An Intel RealSense D435 camera mounted above the gripper provides a close-range eye-in-hand view. The complete operating flow is illustrated in Fig. 2(c). The band servo current serves as the force feedback signal for

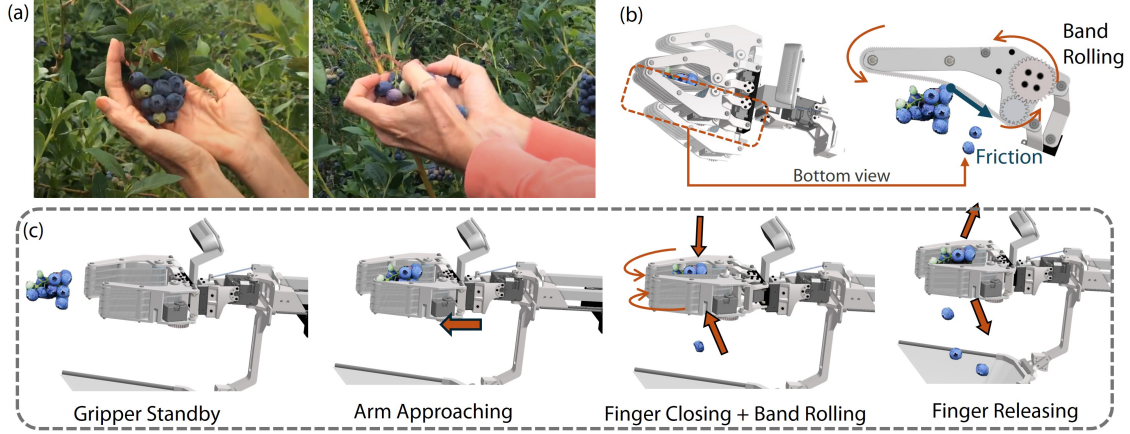


Fig. 2. Illustration of the grasping workflow, finger-closure movements, and harvesting mechanism: (a) inspiration from a human picker’s harvesting demonstration; (b) finger-closure movements and harvesting mechanism; (c) grasping workflow for the gripper design.

the closed-loop regulation of the pulling force, described in Section V.

C. Electronic Architecture

The electronic architecture is shown in Fig. 1(e). Separate voltage rails provide 48 V for stepper motors, 12 V for the Dynamixel XC430 and 2XC430 servos, and 5 V for the Dynamixel XC330 servos. High-level perception and control run on a laptop equipped with an Intel Core i7-10875H CPU operating at 2.30 GHz and an NVIDIA Quadro T1000 GPU. Communication between the laptop, cameras, servo controllers, and microcontroller is established through USB connections. The stepper motors are driven by DM556Y drivers using step-and-direction signals from the Arduino, which also reads the AS5600 joint encoders through an I2C multiplexer and the limit switches through digital inputs.

IV. BLUEBERRY DETECTION DATASET AND MODEL

Cluster detection provides the visual input for subsequent localization and autonomous harvesting, requiring robust detection of small, densely distributed, and frequently occluded blueberries under varying field illumination. To address these conditions, we develop a multi-season field dataset and train a dedicated berry detector for the harvesting pipeline.

A. Dataset Collection and Configuration

Our dataset was collected over four growing seasons (2023–2026) at three research farms across Mississippi and Georgia in the U.S., spanning four rabbiteye cultivars. Images were captured with five cameras of differing focal length, and resolution: three smartphone cameras, a Canon EOS 4000D, and an Intel RealSense D435. The dataset therefore spans variations in season, site, cultivar, camera, and time of day, providing diverse field conditions for detector training and evaluation.

We annotated an instance-segmentation dataset¹ on the Roboflow platform [27] with three ripeness classes: immature, semi-mature, and mature. The semi-mature class was

then merged with the immature class for training because it represented only 1.9% of all annotations and showed ambiguous visual characteristics. Small and densely packed berries, including partially occluded instances, were annotated individually to capture the incomplete appearances commonly encountered in the field. The model was trained and evaluated on a 672-image subset. An additional 40 images were subsequently annotated, resulting in a final open-source dataset of 712 images. A sample gallery and representative annotations are shown in Fig. 3(a) and (b).

B. Image Data Preprocessing and Augmentation

We applied offline augmentation to the training data to improve the detection robustness. Images were auto-oriented to normalize EXIF rotation metadata and resized within 1280×720 px while preserving the aspect ratio. The dataset was divided into training, validation, and test splits of 940, 134, and 68 images, with only the training set expanded through 2× offline augmentation. To account for variations in sensing conditions throughout field operation, the offline augmentation includes exposure changes of $\pm 9\%$, camera-gain variation of 0.05, and additive noise affecting up to 1.09% of pixels. The resulting training set contains 18 994 immature and 17 884 mature berry instances, with the label statistics shown in Fig. 3(c).

C. Detection Model Training Procedure

Using the offline augmented training dataset described above, we fine-tune a COCO-pretrained YOLO26n (2.4M parameters) [28] with an input resolution of 1280 px (`imgsz=1280`). Since individual berries occupy only a small fraction of the image (Fig. 3(c)), a higher resolution than the library default is used to preserve small-target features, while the compact model supports efficient on-board inference.

Unlike the offline augmentation that simulates camera and sensing variations, training-time augmentation primarily modifies scene appearance and is configured to preserve color-encoded ripeness cues and the characteristics of densely packed small targets. Hue jitter is set low (0.015) because ripeness is encoded by color, whereas saturation and brightness

¹Dataset is available at this link.

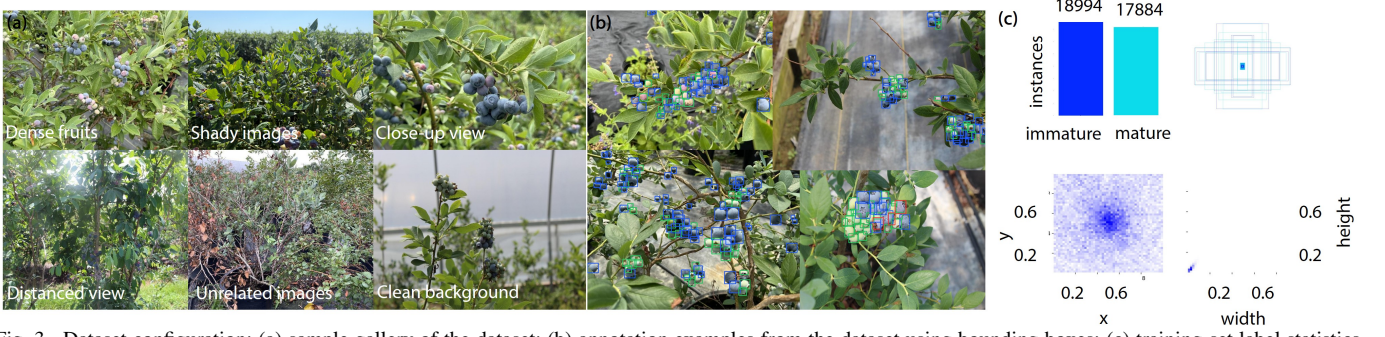


Fig. 3. Dataset configuration: (a) sample gallery of the dataset; (b) annotation examples from the dataset using bounding boxes; (c) training-set label statistics, showing the per-class instance distribution and the normalized bounding-box dimension distribution.

jitter are raised (0.5) to span the illumination range of a harvesting session. Mosaic and MixUp are disabled for the final 15 epochs to allow the model to converge on the natural scale and distribution of densely packed berries.

Because the harvesting controller requires only berry counts and approximate locations, the instance masks were converted to axis-aligned bounding boxes and YOLO26n was trained for object detection. The remaining training settings follow standard practice. Training ran up for 300 epochs with AdamW optimizer with a cosine learning-rate schedule from 10^{-3} to 10^{-5} and early stopping with a patience of 50. Model checkpoints were selected on the validation split, while the test split was held out for final evaluation.

V. AUTONOMOUS HARVESTING CONTROL PIPELINE

In this section, we present the perception, high-level control, and low-level control framework for autonomous cluster-level harvesting, as shown in Fig. 4.

A. Global-to-Local Perception

The perception module for cluster localization operates at two spatial scales. A ZED 2 stereo camera first provides coarse global localization, producing a set of candidate cluster regions $\mathcal{P} = \{\mathbf{p}_i \in \mathbb{R}^3\}$ from the RGB and depth images.

For each candidate region, a scan pose is defined 140 mm away from the region along the horizontal direction toward the robot base, with the camera facing the region. Kinematic

feasibility is evaluated using the inverse kinematics solver described below, with a position tolerance of 5 mm, relaxed to 15 mm if necessary. Candidate poses that meet the position requirement but cannot achieve the required approach orientation are rejected to avoid lateral contact with surrounding branches.

After the arm reaches an accepted scan pose, perception transitions from global localization to close-range cluster scanning using the Intel RealSense D435 on the end-effector. If no cluster is detected over several consecutive frames, the wrist steps through discrete pitch and yaw offsets of up to 40° until one cluster appears in the field of view. Detection bounding boxes predicted by the YOLO26n model are then projected through the depth map to obtain the corresponding 3D cluster point clouds.

These point clouds provide the geometry needed to estimate how each cluster is oriented. The orientation is described by a unit vector $\hat{\mathbf{s}} \in \mathbb{R}^3$, which is inferred from the berry distribution because the thin stem is largely occluded. The cluster point cloud is first filtered using a local K-Nearest Neighbors (KNN) search, and $\hat{\mathbf{s}}$ is then estimated as the leading principal component from Principal Component Analysis (PCA). It is expressed as ${}^T\hat{\mathbf{s}}$ in the tool frame $\{T\}$ and ${}^B\hat{\mathbf{s}}$ in the base frame $\{B\}$.

The estimated cluster position and orientation are combined into a geometric descriptor for subsequent motion planning. Each scan yields N clusters that are tracked across frames within that scan, with the set reinitialized for each new scan.

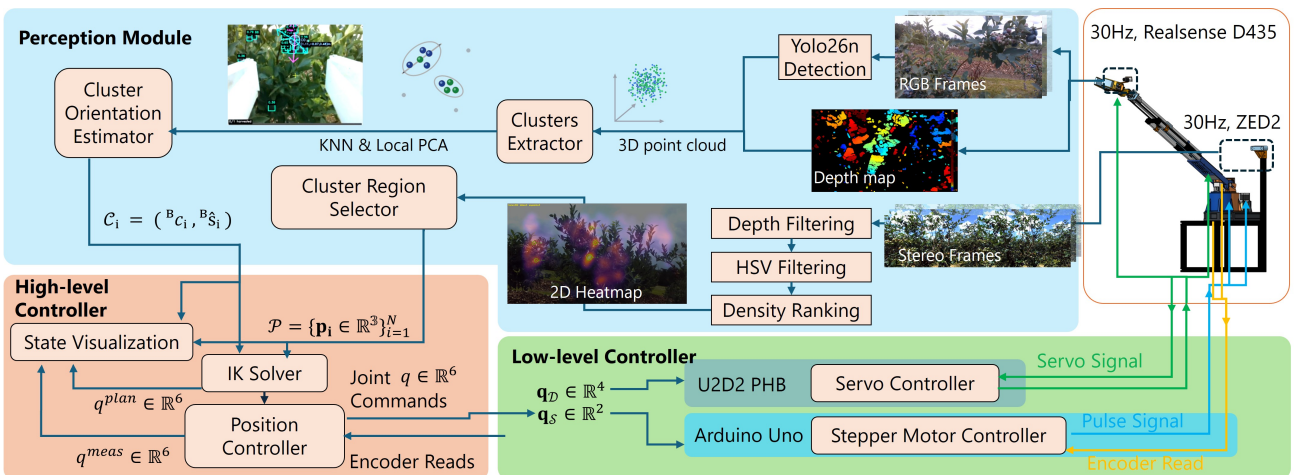


Fig. 4. Blueberry autonomous harvesting pipeline illustration.

The i -th cluster is represented as

$$C_i = ({}^B\mathbf{c}_i, {}^B\hat{\mathbf{s}}_i), \quad i \in \{1, \dots, N\} \quad (1)$$

where ${}^B\mathbf{c}_i \in \mathbb{R}^3$ is the centroid expressed in the robot base frame $\{B\}$, obtained by transforming the final scan-frame estimate using the eye-in-hand transform ${}^B\mathbf{T}_C(\mathbf{q}_{\text{scan}}) \in SE(3)$.

B. Grasp Orientation Planning

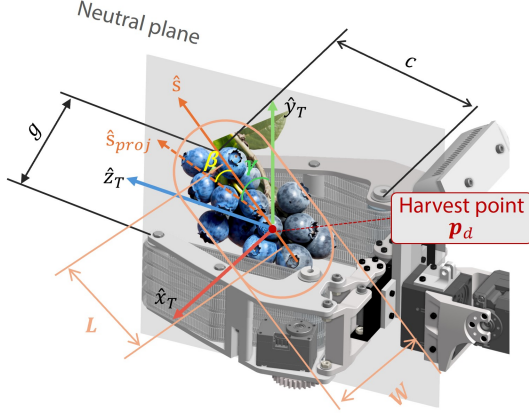


Fig. 5. Tool frame $\{T\}$ and cluster geometry at the harvest point $\mathbf{p}_d$. The fingers open by g along the closure axis $\hat{\mathbf{x}}_T$, the gripper approaches along $\hat{\mathbf{z}}_T$, and the transverse axis $\hat{\mathbf{y}}_T$ spans the contact region between the fingers over an effective band contact length c . The $\hat{\mathbf{y}}_T$ - $\hat{\mathbf{z}}_T$ plane forms the neutral plane. The cluster has stem axis $\hat{\mathbf{s}}$, length L , and width W .

The cluster descriptor specifies the position and orientation of each cluster, but a corresponding gripper orientation must still be determined for harvesting. Grasp orientation planning therefore seeks to align the gripper with the estimated cluster axis $\hat{\mathbf{s}}$. The resulting misalignment is decomposed into two components. The out-of-plane misalignment deviation of the cluster axis is $\beta = \angle(\hat{\mathbf{s}}, \hat{\mathbf{s}}_{\text{proj}})$ and the in-plane misalignment deviation is $\gamma = \angle(\hat{\mathbf{s}}_{\text{proj}}, \hat{\mathbf{y}}_T)$, where $\hat{\mathbf{s}}_{\text{proj}}$ is the projection of $\hat{\mathbf{s}}$ onto the neutral plane (Fig. 5).

Three geometric conditions determine whether a grasp orientation is admissible, using the tool frame and cluster geometry defined in Fig. 5. First, the cluster must fit through the finger opening during insertion, requiring its projected width along the closure axis to satisfy $L \sin \beta + W \leq g$. Second, the two rolling bands must contact opposite sides of the cluster symmetrically to avoid a resulting moment that bends or twists the stem, requiring the cluster axis to lie in the neutral plane at $\beta = 0$. Third, the cluster must remain within the effective rolling region, so its extent along the band-transport direction satisfies $L \sin \gamma + W \leq c$. Berries near the edge of the contact region otherwise experience insufficient rolling distance for reliable detachment.

All three conditions are jointly satisfied at $\beta = \gamma = 0$, corresponding to the ideal configuration in which the cluster axis ${}^T\hat{\mathbf{s}}$ aligns with the transverse tool axis $\hat{\mathbf{y}}_T$, expressed as:

$${}^T\hat{\mathbf{s}} \times \hat{\mathbf{y}}_T = \mathbf{0}. \quad (2)$$

This alignment maximizes the geometric margins for both insertion and rolling contact. As the cluster length L increases, these margins decrease, making longer clusters less tolerant to

orientation errors and motivating cluster-specific orientation adjustment. In practice, however, the achievable adjustment is constrained by the fixed approach direction $\hat{\mathbf{z}}_T$. Wrist roll is therefore used to minimize the deviation from the ideal alignment in (2). Specifically, the roll drives the out-of-plane deviation β to zero, while the remaining in-plane deviation γ is determined by the approach direction. The resulting orientation provides the closest achievable alignment and determines the wrist roll commanded for each cluster.

C. Joint-Space Arm Control

The arm controller takes the cluster descriptor and the planned grasp orientation and produces the joint commands that place the gripper on the target. The controlled point is the tool center point (TCP), located 148 mm distal to the wrist pitch axis (d_6 in Table II). The tool approach vector $\mathbf{a}(\mathbf{q})$ is defined as the tool-frame z -axis, corresponding to the direction along which the gripper engages the cluster.

1) *Inverse Kinematics (IK)*: The proximal azimuth-elevation-extension chain admits a closed-form position solution, but the full chain does not satisfy Pieper's criterion [29]. Because axis 4 is offset by $|a_4| = 24$ mm from the intersection of axes 5 and 6, the wrist is non-spherical and position and orientation remain coupled. We therefore solve the full-chain IK numerically, initialized by the closed-form shoulder solution [30]. For a desired harvest pose, the joint configuration is obtained from the box-constrained nonlinear least-squares problem:

$$\begin{aligned} \min & \|\mathbf{r}(\mathbf{q})\|_2^2 \\ \text{s.t.} \quad & \mathbf{q}_{\min} \leq \mathbf{q} \leq \mathbf{q}_{\max} \end{aligned} \quad (3)$$

where

$$\mathbf{r}(\mathbf{q}) = \begin{bmatrix} (\mathbf{p}_d - \mathbf{p}(\mathbf{q})) / \varepsilon_p \\ \phi(\mathbf{R}_d^T \mathbf{R}(\mathbf{q})) / \varepsilon_R \\ (\mathbf{u}(\mathbf{q}) - \hat{\mathbf{z}}_T) / \varepsilon_u \\ \theta_5 / \varepsilon_\theta \end{bmatrix}. \quad (4)$$

The per-joint limits $\mathbf{q}_{\min}$ and $\mathbf{q}_{\max}$ are defined in Table II, with the inequalities taken component-wise. Here, $\mathbf{p}_d$ and $\mathbf{R}_d = [\hat{\mathbf{x}}_T \ \hat{\mathbf{y}}_T \ \hat{\mathbf{z}}_T]$ denote the desired harvest position and tool orientation, respectively. The approaching axis $\hat{\mathbf{z}}_T$ is selected normal to the estimated cluster axis $\hat{\mathbf{s}}$ and directed from the robot toward the cluster, while the roll about $\hat{\mathbf{z}}_T$ is determined from the cluster orientation as described in Section V-A. The operator $\phi(\cdot)$ maps the rotation error to its axis-angle representation. In addition, $\mathbf{u}(\mathbf{q})$ is the unit vector along $\partial \mathbf{p} / \partial d_3$, representing the instantaneous TCP translation induced by extension of the prismatic joint.

The four residual blocks in $\mathbf{r}(\mathbf{q})$ respectively enforce TCP position, tool orientation, prismatic travel-axis alignment, and a preferred wrist configuration. To reduce collision risk from the lateral sweeping motion of the azimuth joints, the travel-axis term favors configurations in which the final approach is achieved primarily through prismatic extension along the desired approach direction. Aligning $\mathbf{u}(\mathbf{q})$ with $\mathbf{a}_d$ therefore makes the final approach approximately collinear with the tool axis. We then use different scaling parameters for the residual blocks: $\varepsilon_p = 5$ mm, $\varepsilon_R = 10^\circ$, $\varepsilon_u = 5^\circ$. The tighter travel-axis tolerance places greater emphasis on straight prismatic

insertion. The final term uses $\varepsilon_\theta = 33^\circ$ as a weak regularizer that favors an extended wrist when multiple configurations otherwise satisfy the task constraints.

To initialize the numerical IK, the target cluster position is used to evaluate the closed-form solution of the proximal three joints and obtain a nominal configuration. Multiple initial guesses are formed from this nominal solution, a fixed set of seeds spanning several elevation and wrist angles, and a warm start from the current wrist. Each is solved as a box-constrained nonlinear least-squares problem using a trust-region-reflective algorithm. Among the converged solutions, the one with the smallest TCP position error is selected.

2) *Motion Execution*: For each feasible target pose, the IK solver provides a target joint configuration $\mathbf{q}_d$. Motion is executed through a predefined joint sequence to reduce collisions with the surrounding canopy. Before large arm motions, the extension joint retracts to keep the manipulator compact. The azimuth and elevation joints then move toward the target, followed by the wrist joints, and the extension joint extends last to provide an approximately linear approach to the cluster. Joint trajectories are generated by interpolation in joint space, with trapezoidal velocity profiles applied to the azimuth and elevation joints to reduce structural oscillation.

The detected clusters are processed sequentially. For each cluster, the end-effector first moves to its approach pose and then advances to the harvest pose with the planned wrist orientation. After the harvesting action described in Section V-D, the arm returns to the corresponding scan pose before proceeding to the next cluster.

D. Autonomous Gripper Control for Cluster-Level Harvesting

Selective cluster harvesting requires the gripper to operate within the detachment-force window identified in Section VII-B, applying sufficient load to detach mature berries while remaining below the higher detachment force of immature fruit. The gripper therefore regulates the interaction load through motor-current feedback while the rolling bands progressively detach mature berries.

Because the finger motors operate in current-based position control, motor current increases when finger motion is resisted by an enclosed cluster and is therefore used as a surrogate for interaction force. Before harvesting, the unloaded current of each finger is calibrated as μ_i . Initial contact is detected when the window-averaged current exceeds

$$I_{th,i} = \mu_i + I_{cal}, \quad (5)$$

where I_{cal} provides a margin above unloaded motor current and measurement noise caused by gear friction. The current is averaged over a sliding window, so a brief current spike is not detected as contact.

Each finger is independently regulated through its normalized closure $\alpha_i \in [0, 1]$, updated at every control tick as

$$\alpha_i \leftarrow \begin{cases} \alpha_i - \Delta, & I_i > I_{th,i}, \\ \alpha_i + \Delta, & I_i \leq I_{th,i}, \end{cases} \quad (6)$$

where $I_{th,i}$ is the active threshold for the current stage. Thus, each finger closes while the measured load remains below the

TABLE III
GRIPPER CONTROLLER PARAMETERS

Symbol	Value	Description
Δ	0.01	Closure increment per tick
I_{cal}	15 mA	Approach-threshold margin
λ	0.97	Baseline tracking factor (0.67 s)
δ	35 mA	Harvesting-threshold offset
<i>Timing and termination</i>		
Control rate	50 Hz	Closure update rate
Averaging window	12 samples	240 ms
Contact confirmation	0.3 s	Both fingers, to enter harvest
Release margin	8 mA	Above μ_i , for 0.8 s
Vision check	5 s	Interval, max over 8 frames

threshold and opens slightly when the threshold is exceeded, bounding the interaction load. Independent regulation of the two fingers allows the gripper to conform to irregular and off-center clusters. Harvesting begins after bilateral contact is maintained for a sustained interval.

During harvesting, rolling-band motion continuously changes the cluster geometry and corresponding motor load. The fixed contact threshold is therefore replaced by an adaptive threshold. A tracking baseline $\bar{I}_i$, initialized at the onset of harvesting, is updated as

$$\bar{I}_i \leftarrow \lambda \bar{I}_i + (1 - \lambda) I_i, \quad I_{th,i} = \bar{I}_i + \delta, \quad (7)$$

where λ determines the tracking rate and δ defines the allowable current margin above the baseline. The controller therefore responds to transient load increases relative to the slowly varying baseline rather than to the absolute current, allowing the fingers to adapt to the evolving cluster geometry without continuously increasing the compressive load.

The closure controller consequently responds to changes in load relative to the evolving baseline rather than to the absolute current. Because (6) contains no dead band, each finger makes small opening and closing adjustments around the threshold while the bands roll continuously. This maintains contact with the changing cluster geometry while limiting the interaction load as mature berries are progressively detached.

Harvesting terminates based on either current or vision feedback. The current-based condition is satisfied when both finger currents return near their unloaded values μ_i for a sustained interval, indicating that the cluster no longer obstructs finger closure. For vision-based termination, the fingers periodically open and the detector evaluates the remaining mature berries. Harvesting terminates when the detected count falls below a preset threshold, with the maximum count over several consecutive frames used to reduce single-frame misses. The controller parameters are summarized in Table III.

VI. WORKSPACE AND DEXTERITY ANALYSIS

The autonomous harvesting pipeline requires the manipulator not only to reach a target cluster but also to achieve a suitable gripper orientation defined in (2). We therefore analyze both the reachable workspace and the orientation dexterity of the robot. The harvest point is the location at which the gripper contacts the cluster, and the tool frame is attached to

it as shown in Fig. 5. Its reachable set is determined by the kinematic chain of Table II. The extension joint translates the end-effector along the forearm axis, producing a spherical-shell workspace centered at the elevation joint pivot. With the wrist pitch at its neutral configuration, the radial reach spans 937 to 1333 mm, while wrist articulation reduces the minimum reach to 866 mm. Sweeping this region over the azimuth-joint range yields a total workspace of 1.56 m³. Because azimuth rotation does not affect the remaining joint limits, the workspace is rotationally symmetric away from the azimuth boundaries and can be represented by the cross-section (R, Z) , where $R = \sqrt{X^2 + Y^2}$.

At a given harvest position P , multiple joint configurations reach the same point while producing different gripper orientations. We define $\mathcal{Q}(P)$ as the set of all configurations that reach P that satisfy joint limits:

$$\mathcal{Q}(P) = \{ \mathbf{q} : \mathbf{q}_{\min} \leq \mathbf{q} \leq \mathbf{q}_{\max}, \mathbf{p}(\mathbf{q}) = P \}. \quad (8)$$

where $\mathbf{p}(\cdot)$ is the forward kinematics and $\mathbf{q}_{\min}$, $\mathbf{q}_{\max}$ are the joint limits defined in Table II. Each configuration provides a different orientation of the transverse tool axis $\hat{\mathbf{y}}_T$. We therefore define the achievable orientation set $\mathcal{S}(P)$ as all cluster-axis orientations that can be aligned with $\hat{\mathbf{y}}_T$ by at least one configuration in $\mathcal{Q}(P)$:

$$\mathcal{S}(P) = \{ \hat{\mathbf{s}} \in \mathbb{RP}^2 : \exists \mathbf{q} \in \mathcal{Q}(P), \hat{\mathbf{s}} \times \hat{\mathbf{y}}_T(\mathbf{q}) = \mathbf{0} \}. \quad (9)$$

Because the cluster axis is unoriented, $\hat{\mathbf{s}}$ and $-\hat{\mathbf{s}}$ are equivalent, giving a total orientation-space measure of 2π steradians. The orientation dexterity $\rho(P)$ is defined as the fraction of all cluster-axis directions contained in $\mathcal{S}(P)$,

$$\rho(P) = \frac{1}{2\pi} |\mathcal{S}(P)| \in [0, 1], \quad (10)$$

The achievable orientation set is evaluated directly from forward kinematics. We uniformly sample 4×10^3 cluster-axis directions over $\mathbb{RP}^2$ and estimate $\rho(P)$ as the fraction of directions that are achievable at P . The metric is evaluated on a 25-mm grid throughout the reachable workspace. At this grid resolution, the numerical uncertainty in ρ is bounded by ± 1.3 percentage points at the median and ± 2.8 percentage points in the worst case.

As shown in Fig. 6, the orientation dexterity remains largely uniform throughout the reachable workspace. The volume-weighted median is 31.2% with an interquartile range of 27.3% to 31.8%, and 89% of the reachable volume has $\rho > 20\%$. Lower dexterity is concentrated near two workspace boundaries, where joint limits reduce the range of configurations available at a given position. Near the inner radial boundary, prismatic retraction limits elevation joint variation, yielding an average dexterity of 25.5% between 866 and 1000 mm. Below 400 mm, the elevation joint limit reduces the average dexterity to 23.1%.

VII. EXPERIMENTS AND RESULTS

A. Perception Test Results

As mentioned in Section. IV, the detection model need to provide accurate and robust berry detection under field

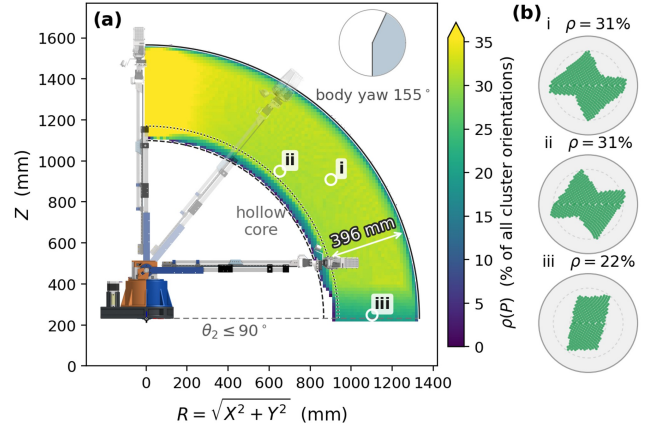


Fig. 6. Workspace and grasp-orientation dexterity of the harvest point, evaluated on the (R, Z) section: (a) reachable envelope, colored by the orientation dexterity $\rho(P)$ of (10), the share of all cluster orientations that can be grasped exactly at that position; (b) the achievable orientation set $\mathcal{S}(P)$ at the three marked positions, on a Lambert equal-area chart of $\mathbb{RP}^2$ centered on the gripper axis at the neutral wrist. Each disc maps the space of cluster-axis directions onto a plane, with directions parallel to the gripper axis at the center and perpendicular ones at the outer edge. The chart is equal-area, so the green share of each disc is exactly $\rho(P)$.

TABLE IV
DETECTION PERFORMANCE ON THE HELD-OUT TEST SPLIT

Class	Inst.	P	R	F_1	mAP ₅₀	mAP ₅₀₋₉₅
Immature	1665	0.799	0.649	0.716	0.742	0.478
Mature	1134	0.795	0.699	0.744	0.792	0.574
All	2799	0.797	0.674	0.730	0.767	0.526

Evaluated at 1280 px without test-time augmentation.

conditions. We therefore evaluated the model on the test split, which contains 68 images with 2,799 annotated instances, with an average of 41 berries per image. Our detection model reaches an mAP₅₀ of 0.767 and a stricter mAP₅₀₋₉₅ of 0.526, averaged across both classes. At the operating point that maximizes the F_1 score, a metric that balances precision and recall, the model achieves a precision of 0.797 and a recall of 0.674 Table IV). The corresponding precision-recall curves are shown in Fig. 7(a). Among these metrics, mAP₅₀ is particularly relevant to harvesting because the controller relies more on correctly detecting and counting berries than on highly precise bounding-box localization.

The mature class achieves higher recall and mAP than the immature class, while maintaining comparable precision (Table IV). This higher detection performance for mature fruit likely results from their more distinctive and consistent appearance, whereas the immature class also includes partially ripened berries with greater visual variation. This is particularly important for harvesting because both target selection and the decision to terminate harvesting a cluster depend on the count of mature berries.

The normalized confusion matrix of Fig. 7(b) distinguishes between two types of detection errors. Misclassification of maturity is rare, with 3% of immature berries predicted as mature and 3% of mature berries predicted as immature. The

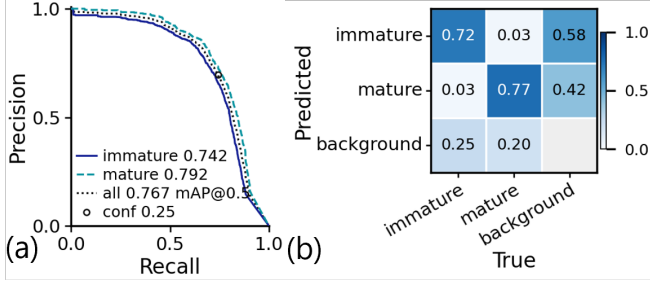


Fig. 7. Detection model evaluation on the held-out test split: (a) precision–recall curves at an IoU threshold of 0.5 for the immature and mature classes and their average; (b) confusion matrix, normalized so that each ground-truth column sums to one; columns are ground-truth classes and rows are predictions.

dominant error is the missed detections, with 20 % mature berries and 25 % immature berries assigned to the background.

Missed detections matter more than misclassification for harvesting. A single observation may under-count the remaining mature berries and terminate harvesting prematurely. The controller therefore aggregates detections over multiple frames during the inspection window (Section V-D), using the maximum observed count over the window rather than a single frame. Berries that remains occluded from the camera throughout the inspection window, however, cannot be recovered by this strategy.

B. Gripper Force Test

1) *Experiment Setup*: Selective harvesting requires the gripper to generate a pulling force sufficient to detach mature berries while remaining below the detachment force of immature fruit. We therefore characterized both the pulling force generated by the gripper and the berry detachment force across maturity stages, using the setups shown in Fig. 8(a) and (b). Both experiments used an ATI Nano17 six-axis force sensor sampled at 100 Hz, with F_z aligned with the harvesting direction and used for force analysis.

The pulling force transmitted by the gripper was characterized indoors using the control algorithm described in Section V-D. A spherical object was coupled directly to the force sensor, and three diameters were tested in turn, 23 mm, 30 mm and 35 mm. For each diameter, we varied the servo current offset I from 5 mA to 50 mA in step of 5 mA. With two to four repetitions per setpoint, the variation comprises 79 trials, with 25 s to 45 s of force data recorded per trial. The reported pulling force was obtained as the peak of a 50 ms moving average of F_z , suppressing single-sample transients while preserving the physical force peak. Because the two fingers contact the whole cluster on opposite sides, the measured force was divided by two to estimate the equivalent single-side-contact force applied to an individual berry.

The detachment force was measured in the field across maturity stages. The sensor was connected to a berry clipper with a soft contact surface. After clipping a berry, the sensor was drawn along the harvesting axis until the berry separated from its stem, and the peak of F_z was recorded as the detachment force.

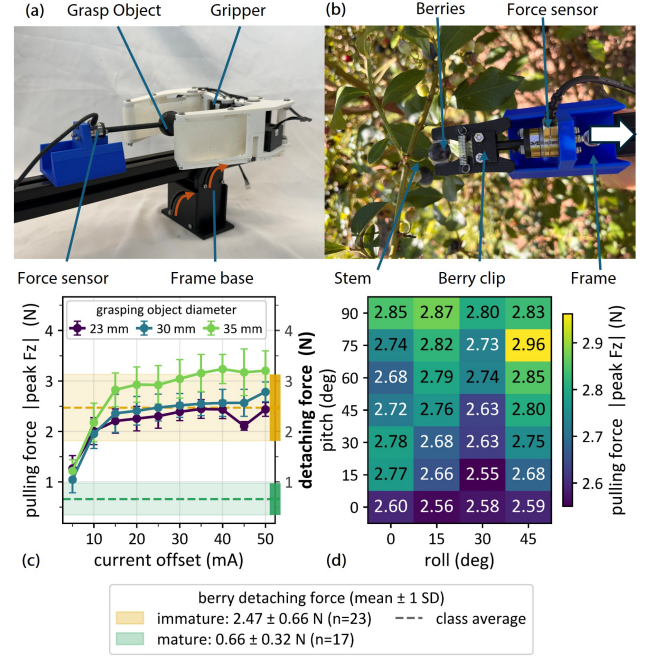


Fig. 8. Experiment setup and gripper force characterization: (a) gripper pulling-force measurement and (b) blueberry detaching-force measurement; (c) pulling force ($|peak F_z|$) versus servo current offset for three grasping-object diameters (left axis), with shaded bands marking the measured berry detaching force (right axis, mean ± 1 SD) for immature and mature fruit; (d) pulling force over wrist pitch and roll at a fixed 30 mA offset.

2) *Experiment Results*: Selective pick requires that the pulling force be set finely and predictably through the servo current offset I and that, for a given setpoint, the delivered force be repeatable.

As shown in Fig. 8(c) the pulling force increases smoothly and monotonically with current, with slopes of 0.09 N mA^{-1} to 0.16 N mA^{-1} across the three object diameters. The force also increases with object diameter across the tested range of 23 mm to 35 mm. These trends enable the pulling force to be regulated through the commanded current according to the grasped object size. At higher currents, the force gradually saturates beyond approximately 15 mA, bounding the load transmitted by the gripper and thereby limiting over-gripping.

At a fixed setpoint, the pulling force varied only 1.4 % to 3.7 % on average across repeated grasps (worst case 7.2 %), confirming repeatability of the commanded force. The force control also remained robust to wrist pose, with the pulling force varying by only 0.41 N (about 15 % of the mean) across the full pitch and roll grid in Fig. 8(d). Together, these results indicate that the gripper can reproducibly regulate the pulling force without requiring precise end-effector orientation.

The horizontal dashed lines and shaded bands in Fig. 8(c) show the actual in-field detachment-force measurements for mature and immature berries. Mature berries detached at $(0.66 \pm 0.32) \text{ N}$ and immature berries at $(2.47 \pm 0.66) \text{ N}$, a ratio of 3.7 in mean detaching force with no overlap at one standard deviation. The intervening 0.8 N window allows the pulling force to detach ripe fruit while leaving unripe fruit attached. Because the pulling force is reproduced to within a few percent, the controller can maintain an operating

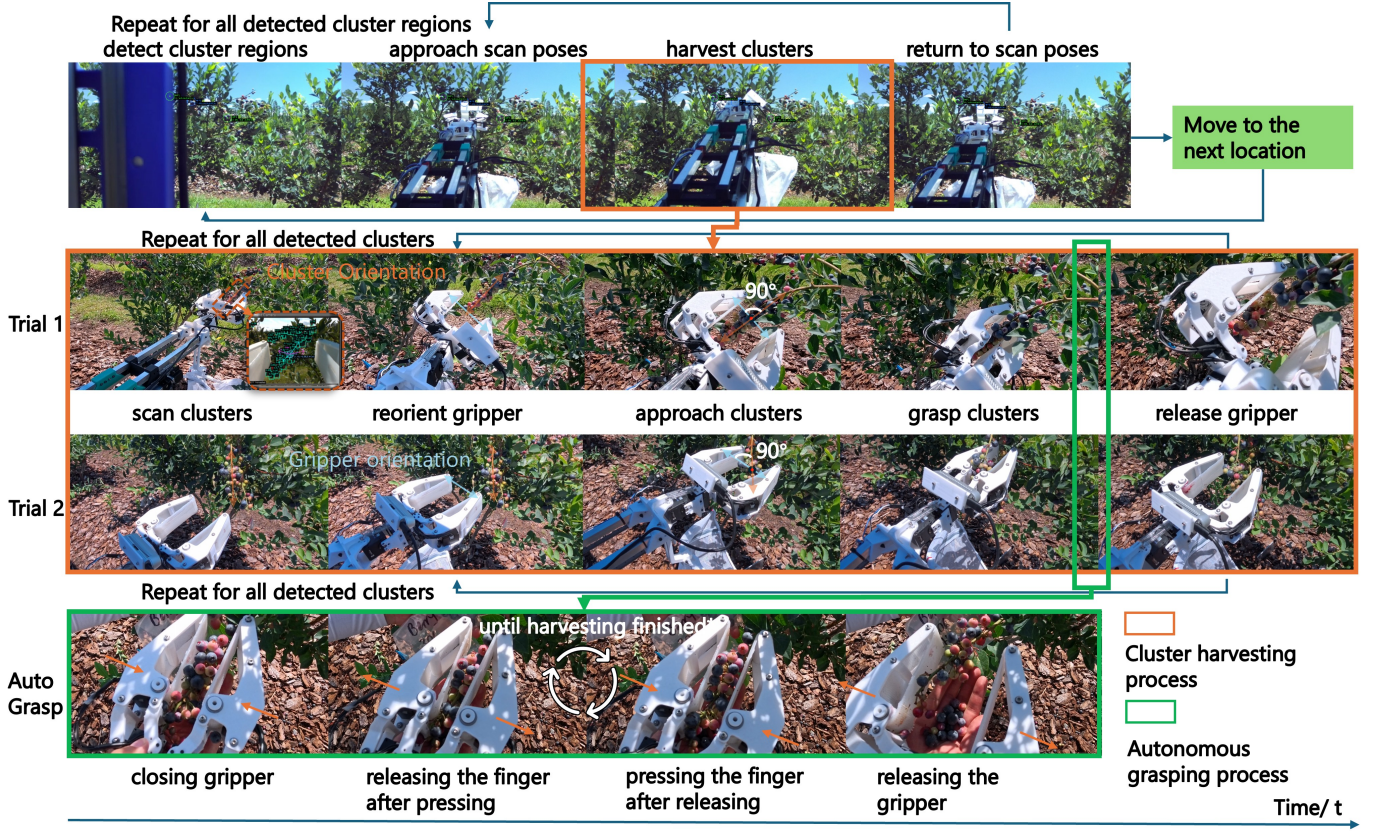


Fig. 9. Field demonstration of the autonomous harvesting workflow, read left to right as a time sequence across three nested levels. The platform services each detected cluster regions (top), the arm navigates to scan pose and detect clusters before approaching and grasping it (middle, orange), and the gripper repeats closing and opening cycles until the termination criterion is met (bottom, green).

point within this selective detachment window across repeated grasps. The achievable force range also depends on object size. The saturation force increases with object diameter, from 2.34 N at 23 mm to 3.09 N at 35 mm. The current setpoint can therefore be adjusted according to the measured cluster

C. In-Field Test Results

System performance was evaluated through field experiments characterizing the crop environment, harvesting performance, post-harvest berry firmness, and gripper throughput.

1) *Field Test Setup*: Field tests were conducted in a blueberry research farm on three cultivars: Alapaha (southern highbush), Brightwell, and Vernon (rabbiteye). The robot was mounted on a mobile platform (Rover Robotics MAX) for teleoperated in-field transport. Autonomous harvesting tests were then conducted at several fixed locations throughout the day under naturally varying illumination conditions. Human hand picking was also performed as a baseline for comparison. Harvested berries from both robotic and manual picking were subsequently evaluated using a FruitFirm 1000 (CVM, Inc.) to compare post-harvest firmness. In addition, 72 clusters were analyzed to characterize the canopy architecture in terms of canopy occlusion and cluster density. The autonomous harvesting performance was evaluated over 25 attempted clusters.

2) *Canopy architecture characteristics*: The field environment characterization quantifies the visual challenges faced by the perception pipeline. Only 40.3% (29 of 72 analyzed

clusters) of the clusters surveyed lay within the field of view of the global camera, with the remainder severely occluded by leaves and stems. Therefore, this limited observability from a fixed viewpoint demonstrates the need for the global-to-local perception strategy.

3) *Harvesting Success, Efficiency, and Fruit Quality*: The harvesting trial measures both end-to-end and modular performance of the full pipeline. Of 25 harvesting attempts, 23 succeeded and 2 failed, giving a success rate of 92%. In both failures, the arm could not bring the gripper onto the cluster, so the fruit was never engaged. Performance is therefore limited by perception and approaching accuracy rather than by the harvesting mechanism, which is consistent with the occlusion rate reported above. With 25 attempts, the result demonstrates the feasibility of the integrated system but should not be interpreted as a comprehensive reliability estimate. Two representative grasping trials are shown in Fig. 9.

Gripper efficiency for cluster-level harvesting was evaluated independently by manually aligning the gripper with each cluster before harvesting. The gripper achieved 32 berries per minute against 55 for a human picker. This isolates the gripper from the autonomous pipeline and shows that the gripper itself achieves 58% of the manual harvesting rate.

As another important aspect of harvest performance, we measured the firmness of individual berries after field tests. Berry firmness provides an indicator of post-harvest fruit quality, as mechanical damage and bruising generally reduce tissue firmness. The autonomously harvested fruit measured

166.22 g mm⁻¹ against 195.59 g mm⁻¹ for the hand-picked reference, corresponding to a 15 % reduction. Two effects contribute to this gap. The first is mechanical loading during grasping. The second is the longer interval between autonomous harvesting and post-harvesting measuring experiments.

Because both factors can reduce measured firmness, the observed 15 % difference represents an upper bound on the reduction attributable to the harvesting mechanism itself. Section VIII examines the two effects separately. Even under this conservative interpretation, the observed reduction compares favorably with the 16 % to 26 % severe bruising rates reported for mechanical harvesters in Section I.

VIII. CONCLUSIONS AND DISCUSSION

In this paper, we presented CLASP, an autonomous robotic system for selective cluster-level blueberry harvesting through force-regulated interaction. We use ripeness-dependent detachment-force characterization to establish the mechanical basis for selective harvesting, while the soft rolling-band gripper enabled force regulation without dedicated force sensing or per-berry pose estimation. The integrated system achieved a 92 % autonomous harvesting success rate in field trials. In addition, the accompanying multi-season dataset supports detector development across diverse canopy and lighting conditions.

Several opportunities remain for further improving and validating the system. First, autonomous throughput can be increased by optimizing arm trajectories and reducing transit time between harvesting poses. Under manually aligned grasp poses, the gripper already achieved 58 % of the manual picking rate, indicating that system-level motion rather than the detachment mechanism itself is a major contributor to cycle time. Second, the current field study evaluated 25 clusters at a single site, and future multi-site experiments will further characterize performance across cultivars, canopy structures, and environmental conditions. Third, autonomously harvested fruit showed a 15 % reduction in firmness relative to hand-picked fruit. Because the autonomous samples experienced a longer interval before measurement, future time-matched experiments will more precisely distinguish harvesting-induced effects from post-harvest storage and thermal exposure.

Future work will focus on further expanding the multi-season dataset, collision-aware trajectory planning for more reliable and efficient harvesting, and coordinated multi-arm operation to increase system-level throughput.

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